MIL-C-60184(MU)
AMENDMENT 9
23 November 1970
SUPERSEDING
AMENDMENT 8
23 October 1969

MILITARY SPECIFICATION

COMPUTER SET, GUN DIRECTION: 10525735

This amendment forms a part of Military Specification MIL-C-60184(MU), dated 18 March 1965.

* Paragraph 2.1, under DRAWINGS: Delete "F10526109 Indicator Multilayer Panel: 10526109; IEL 10525735 List of Inspection Documents for Computer Gun Direction: M18".

* Paragraph 3.3: Delete "e. Indicator Multilayer Panel: 10526109 - F10526109".

Paragraph 3.5.2: Delete the first sentence and substitute the following:

"The computer while in its field case, shall meet the requirements of 3.7.1 (the only tests to be performed utilized Connector J013 to perform the output voltage measurements of Table I at a nominal 120 VAC line to neutral input voltage and the kickout tests, 4.5.3(d), of Table VIII), and 3.7.3 through 3.7.9. These tests are to be performed at an ambient temperature of 75 ± 15°F while the computer is subjected for 6 hours to an atmosphere of sand and dust."

Paragraph 3.5.3: Change subparagraph "3.4.3.1" to "3.5.3.1".

Paragraph 3.5.3.1b: Delete and substitute:

"b. The temperature warning indicator lamp shall energize at an ambient temperature greater than 117°F or an internal temperature lower than -18°F."

Paragraph 3.5.3.1c: Delete and substitute:

"c. The computer shall de-energize at an ambient temperature greater than 125°F or an internal temperature lower than -40°F."

Add the following new subparagraph to 3.5.3.1:

"d. The computer may de-energize after sustained operation (sustained operation is defined as, operation of at least 1 hour) at an ambient temperature of +125°F. This is not to be considered a failure. In order to achieve sustained operation after this occurrence, the chamber temperature will be reduced, in increments of 2°F, to the point where operation may be sustained, but not less than +121°F."

FSC 1220

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Paragraph 3.7.1:

(a) Change "132 volts rms and above" to "132 ± 4 volts rms".

(b) Change "108 volts rms and above" to "106 ± 4 volts rms".

(c) Change "140 volts rms and above" to "140 ± 4 volts rms".

(d) Change "100 volts rms and above" to "98 ± 4 volts rms".

(e) Change "80 volts rms and above" to "82 ± 4 volts rms".

Add the following subparagraph to requirement 3.7.1:

"e. The power ready indicator which illuminates when the computer is energized shall blink continuously when the blower cut-off switch is actuated manually and the marginal test switch is in the "OFF" position. This condition shall also exist when the rear cover is installed as the cover automatically actuates the blower cut-off switch."

Delete the second sentence of 3.7.7 and substitute the following:

"The delay shall be minimized such that on either (a) rapid entry (maximum of 0.8 second time interval between key depressions) of any digit on the keyboard, the nixie indicator shall light before the next depression of a key or (b) using a routine which causes the nixie display of a character or characters to shift at a rate equivalent to keyboard entry (maximum of 0.8 second time interval between shifts of nixie display), the display shall shift at a minimum rate of 1.25 nixie positions per second."

* Paragraph 3.8: Change references to "70 hour run-in" to "50 hour run-in"; delete "and 3.8.1" in second sentence and delete third sentence. Fourth sentence: Change "3.7.3 through 3.7.8" to "3.7.8 and 3.7.9".

* Paragraph 3.8.1: Delete in its entirety and substitute:

"3.8.1 Vibration.- The computer shall be capable of withstanding a simple harmonic motion with a single amplitude of 0.015 inch (0.03 inch total excursion) and a frequency varying uniformly between the limits of 10 and 55 cycles per second and return to 10 cycles per second in approximately 1 minute. The computer shall be capable of withstanding this motion for a period of 10 minutes in each of the 3 mutually perpendicular axes (total 30 minutes)."

* TABLE A: Delete Characteristics 133 and 134.

Paragraph 4.3.2: Delete and substitute:

"4.3.1 General.- One unit shall be selected at random as a standard sample from each 50 produced. The sample shall meet the requirements and tests in Table B and paragraph 4.2.1.1."

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Delete the third sentence of 4.5.1.2 and substitute the following:

"During the operating portion of the test, defined in 4.5.2, 4.5.3 (only tests of 4.5.3 to be performed are those voltage measurements of Table I, whose outputs are available at Connector J013, using a nominal 120 VAC line to neutral input voltage and the kickout tests of Table VIII, outputs also available at J013), and 4.5.5 to 4.5.11, inclusive, the computer air filters shall be replaced as required to obtain proper operation."

* Paragraph 4.4.2.1: Delete in its entirety.

Paragraph 4.5.1.6: Delete in its entirety and substitute:

"4.5.1.6 Salt spray.- This test shall be conducted in accordance with the salt spray test defined in Method 101, Condition B of MIL-STD-202 using a 5 percent salt solution. During exposure, the computer rear field case cover will be installed and the front cover removed at the end of the exposure period (48 hours). Salt deposits shall be removed from all exposed surfaces by a gentle wash, not exceeding 30 minutes, in running water not warmer than 100°F, using a soft-hair brush or plastic-bristle brush. The computer shall then be air dried in a controlled environment at 72°F ± 5°F and a relative humidity of 30 to 50 percent for a period of 24 hours. Subsequent to the salt spray test, the computer shall be visually examined for physical damage then tested as outlined in 4.5.5 through 4.5.11 inclusive."

Paragraph 4.5.1.8, first sentence: Delete "with all covers in place." Second sentence: Delete "of the front cover". Fourth sentence: Delete "covers shall be removed".

Paragraph 4.5.3(b): Insert the following sentence between the first and the second sentence: "Connect the capacitors of Column F of Table VIII to the indicated supplies in the computer frame; the capacitors shall be properly polarized with the supply."

Paragraph 4.5.3(c): Insert the following sentence between the first and second sentence: "The capacitors connected in subparagraph "b" above shall remain connected."

Paragraph 4.5.3(d): Delete "+.5 ± .5 volts" and substitute "0 to +2 volts".

Paragraph 4.5.3.1: Insert the following sentence between the first and second sentences: "The capacitors in 4.5.3(b) shall remain connected."

Paragraph 4.5.3.1: Add new subparagraph:

"(c) Energize computer and observe that when the power ready light comes on it shall not blink. Manually actuate the blower cut-off switch to

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cause the power ready light to blink continuously. If the blower cut-off switch is actuated while the computer is energized, the power ready light may be de-energized indicating the computer has entered on out of sync mode. This is considered a normal condition and is not to be considered a computer malfunction."

Paragraph 4.5.4: Add the following:

"A maximum of 20 readings checked during the following tests of this paragraph may fall outside the specified limits within the limitations listed below:

a.	4.5.4.3(f)	65 ± 10%
b.	4.5.4.4(e)	8.6 volts
c.	4.5.4.5(d)	.55 volts
d.	4.5.4.6(d)	.55 volts"

Paragraph 4.5.4.3(f), last sentence: Delete "75" and substitute "65".

Paragraph 4.5.4.3(h): Delete "1%" and substitute "5%".

Paragraph 4.5.9.5(b): Delete in its entirety and substitute:

"(b) The signal at Pin W on J10 shall be equivalent to the signal shown in Figure 2(c). The information signal for a binary "1" shown in Figure 2(b), shows the correspondence between the information signal and the strobe signal. Information which appears on the data lines at times when the strobe is not true (true is the -4.0 ± 2.0 volt level shown in Figure 2) is to be disregarded."

Paragraph 4.5.9.5: Add subparagraphs (c) and (d):

"(c) Generate and execute a routine to input field data code".

"(d) The signal at Pin Y on J17 shall be equivalent to the signal shown in Figure 2(a)".

Paragraph 4.5.9.9: Delete the second and third sentences and substitute the following:

"The routine shall either (a) cause the nixie indicators to display each digit entered from the keyboard before the depression of the next key when entry is made at a minimum rate of 1.25 digits per second, or (b) cause the nixie display of a character or characters to shift at a rate equivalent to keyboard entry (minimum shift rate of 1.25 nixie positions per second) for compliance with 3.7.7."

Paragraph 4.5.10(a): Add "Rev A" after "8213315-17".

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Paragraph 4.5.10(b): Change "9 00000 00000 05055" to "9 00000 00001 05055".

Paragraph 4.5.10.1: Delete Matrix selection "F-5" and substitute "E-2".

Paragraph 4.5.10.1(e): Delete in its entirety and substitute the following:

"(e) Depress set-up button. Using targets 1 through 5 of Table XXXI, enter the problems of Table XXXII.

Enter:

Battery A	in	Marginal 5
Battery B	in	Marginal 4
Battery C	in	Marginal 3
Battery D	in	Marginal 2
Battery E	in	Marginal 1

Select matrix position H-G. Depress SM button and then depress "O". Obtain appropriate solutions of Table XXXIII."

* Delete paragraphs "4.5.12 through 4.5.12.2.1" and substitute:

"4.5.12 Run-in.- As specified in 3.8 and after each computer has successfully passed all the performance tests, each computer shall be run for a total of 50 hours excluding down-time. The tests shall be run for at least 4 hours at -22°F, 4 hours at +125°F and at least 30 hours after the -22°F, +125°F tests are completed.

4.5.12.1 Vibration run-in.- The computer shall be tested as outlined in MIL-F-13926, Procedure II, Condition B except the frequency, amplitude, direction and duration shall be as specified in 3.8.1. Subsequent to this test, the computer shall be visually inspected for physical damage and shall be operated for at least 4 hours, excluding down-time. Vibration time shall not be part of run-in time. During the operation, the following test shall be successfully performed with the computer's front and back covers removed:

a. The memory shall be cleared by writing a 1 in a field of zeros (00000100000) into all memory locations.

b. Repeat 4.5.10 through 4.5.10.3.1(e) and 4.5.11.

4.5.12.2 High temperature run-in.- The computer shall be placed in a chamber and the temperature changed at a rate not to exceed 2° per minute, until +125°F is reached. After stabilization of +125°F, the computer shall

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be operated for at least 4 hours excluding down-time and stabilization time. During this operation, the following tests shall be successfully performed with the computers' front and back covers removed.

a. The memory shall be cleared by writing a 1 in a field of zeros (00000100000) into all memory locations.

b. Repeat 4.5.10 through 4.5.10.3.1(e). and 4.5.11.

c. Two (2) hours of computer operation shall be in its normal mode specified in 4.5.2 and for 2 hours the computer shall be on its side (rotated 90°)."

* Paragraph 4.5.12.4: Delete and substitute:

"4.5.12.4 The remainder of the 50 hours, the computer shall be operating at temperatures between 55°F to 95°F. The tests specified in 4.5.10 through 4.5.10.3.1(e) and 4.5.11 shall be performed."

* Table I, Column B**, Lines 1, 2, 3 and 4: Change "50" to "25"; line 8: Change"100" to "25"; line 10: Change "50" to "25". Column A, line 14: Change "-25.0 ± 1.08*" to "-25.7 ± 1.6*"; line 15: Change "-30.5 ± 1.28*" to "-30.5 ± 1.83*". Add the following sentence to the second paragraph under Table I: "Ripple voltage measurements shall also exclude noise less than 18 usec and less than 155 mv."

Table VIII, add new Column F as shown below:

"Item	F Load Capacitors
1	22uf, 15v ± 20%
2	
3	56uf, 6v ± 10%
4	100uf, 25v ± 20%
5	
6	
7	600uf, 3v ± 10%
8	1200uf, 3v ± 20%
9	
10	
11	300uf, 15v ± 20%
12	
13	
14	
15	
16	
17	
18	

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Table IX, Column A: Delete and substitute the following:

"A
Line Voltage
± volts rms
140
98
82
106
132"

Figure 1: Delete lower limit range "-5.4v" to "-6.6v" and substitute "-3.7 ± 0.8v".

Figure 2, Note A: Delete "not less than 2 microsec"; Note B: Delete "not less than 2 microsec" and substitute "not more than 2 microseconds"; Note C: Delete "nor more than 2 microsec" and substitute "or more than 3 microseconds".

Tables XXXI, XXXII, XXXIII and XXXIV: Delete and substitute:

"TABLE XXXI

TARGET INFORMATION

Bat.	Target	Target East	Target North	Target Alt
A	1	33125	00768	428
A	6	36500	01100	450
B	2	91235	11563	27
B	7	82000	08500	75
C	3	11230	07178	105
C	8	09800	07000	140
D	4	99157	67724	105
D	9	97000	58000	130
E	5	50889	31330	389
E	10	53000	31500	410

TABLE XXXII

Lat N 50
GRID DECL +50
PROBLEM INFORMATION

Battery	A-105	B-155	C-105	D-155	E-105
Bat. East	33485	86240	12340	00294	50000
Bat. North	99567	10070	08560	56767	30000
Altitude	422	20	1000	105	500
Bat. AZ	1600	4800	3200	1600	1600
Bat. Defl.	2800	2600	2800	2600	2800

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TABLE XXXII (Cont'd)

*Defl. Corr.	L1	R10	R20	R10	R15
*Time Corr.	-.2	+.5	+.5	+.5	+.6
*Range Corr.	-50	+50	+50	+100	-100
Charge	1	3	1	1	1
Proj. Type	3	2	4	6	6
Fuse Type	2	2	7	6	6

*Charge Associated

TABLE XXXIII

PROBLEM SOLUTIONS
MET. STANDARD

Bat.	Target	Deflection Low	Deflection High	Fuse Time Low	Fuse Time High	Quad. Elev. Low	Quad. Elev. High
A	1	1500	1701	4.4	39.0	170	1401 NSL
A	6	94	105	20.0	27.0	610	845
B	2	6098	6146	26.8	46.6	520	1083
B	7	2961	3028	22.1	48.2	430	1172
C	3	2094	2223	11.3	35.1	220	1288 NSL
C	8	----	1794	----	29.3	0	930 NSL
D	4	----	----	----	----	0	....0 NSL
D	9	5442	5484	21.5	35.1	644	1074
E	5	589	746	6.6	37.8	267	1346 NSL
E	10	69	103	18.8	28.7	534	902

TABLE XXXIV

PROBLEM SOLUTIONS
MET "H" TAPE

Bat.	Target	Deflection Low	Deflection High	Fuse Time Low	Fuse Time High	Quad. Elev. Low	Quad. Elev. High
A	1	1499	1691	4.4	39.0	170	1397 NSL
A	6	99	108	20.4	26.0	623	807
B	2	6103	6155	27.2	46.7	529	1090
B	7	2957	3018	21.9	48.3	426	1173
C	3	2093	2209	11.2	35.2	217	1290 NSL
C	8	1753	1787	24.1	29.5	694	937
D	4	----	----	----	----	0	0
D	9	5422	5466	21.5	35.1	645	1073
E	5	601	755	7.4	37.8	276	1338 NSL
E	10	88	114	19.7	27.7	561	861"

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Paragraph 30.3.1: Delete second sentence and substitute:

"Information is loaded in octal Teletype or Field data code."

Paragraph 30.3.1(c): Insert the following sentence between the first and second sentence:

"In initial program fill the last character entered is normally a HALT code".. Change last sentence to read: "then the COMPUTER code will start computation at the address specified in L when the computer leaves the manual HALT mode."

Paragraph 30.3.2.1(c)(6): Delete "Manual" and substitute "Program".

Paragraph 30.3.2.3(a): Add "by character".

Paragraph 30.3.2.3(b): Add "by word".

Paragraph 30.3.2.3(c): Add "by character".

Paragraph 30.3.2.3(d): Add "by word".

Paragraph 30.4.1: Delete "ENTER" and substitute "INITIATE".

Paragraph 30.4.2.3(a): Delete "8-bit Character" and substitute "alphanumeric".

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